Grade 11 – Term 1
C4 Practice Activity 4 with Solutions

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Evaluated criterion

- C4: Applies the exponential distribution to constant-rate waiting-time situations and assesses its suitability.

Introduction

The exponential distribution is commonly used to model the time T until the next occurrence of an event. In general, we write

$$T \sim \text{Exponential}(\lambda), \qquad \lambda > 0,$$

where λ is the rate at which events occur.

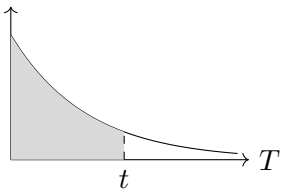
For an exponential random variable, the possible waiting times satisfy

$$T \geq 0.$$

The general case can be summarized by the following relationships:

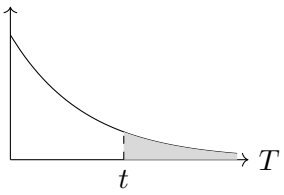
Concept	General expression
Random variable	$T =$ waiting time until the next event
Possible values	$T \geq 0$
Rate	$\lambda > 0$

The three basic probability cases are shown below. The equation is presented alongside each graph, and the shaded region represents the corresponding probability.



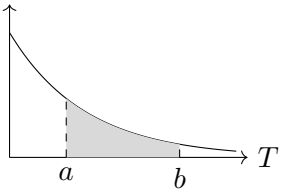
$P(T < t) = 1 - e^{-\lambda t}$

Less than



$P(T > t) = e^{-\lambda t}$

Greater than



$P(a < T < b) = e^{-\lambda a} - e^{-\lambda b}$

Between

These relationships allow us to move from a specific context to the general exponential model. In the following problems, we will identify the random variable, determine the rate parameter when necessary, and use the appropriate probability expression.

1 Identifying the Random Variable

Problems:

1.1. Customer-service calls
1.2. Food-delivery requests
1.3. Bank security alerts
1.4. Battery life of mobile devices

1.1 Mobile-data support requests

A mobile phone company provides technical support for customers who experience problems with their mobile data. Throughout the afternoon, customers call the company's support line to report problems or ask for assistance. The company wants to understand how frequently these calls arrive.

To study this, an analyst randomly chooses a moment during the afternoon and starts a timer. The analyst then records how much time passes before the next mobile-data support call arrives. For example, if the analyst starts the timer at 2:15 p.m. and the next call arrives at 2:18 p.m., the recorded time is 3 minutes.

Question.

C4.1 Identify the random variable and state the possible values it can take.

C4 - Identify the random variable and its possible values

Let

$T =$ the time from a randomly selected observation time until the next mobile-data support call arrives.

The random variable T measures elapsed time, so it cannot take negative values. Therefore,

$$T \geq 0 .$$

1.2 Food-delivery requests

A food-delivery platform receives orders from customers throughout the day. Delivery requests can arrive at any time, and the company wants to understand how much time typically passes between consecutive requests in a neighbourhood.

To study this, an analyst focuses on one neighbourhood and observes the platform after a delivery request has been received. The analyst starts a timer and records how much time passes before the next delivery request from that neighbourhood arrives. For example, if a request is received at 6:20 p.m. and the next request arrives at 6:24 p.m., the recorded time is 4 minutes.

Question.

C4.1 Identify the random variable and state the possible values it can take.

C4.1 - Identify the random variable and its possible values

Let

$$T = \text{the time from one delivery request until the next request arrives}$$

The random variable T measures elapsed time between two consecutive requests, so it cannot take negative values. Therefore,

$$T \geq 0 \ .$$

1.3 Bank security alerts

A Colombian digital bank continuously monitors its customers' transactions for unusual activity. When a transaction meets certain risk criteria, the bank's monitoring system automatically generates a security alert. The bank wants to understand how much time typically passes between these alerts.

To study this, an analyst randomly chooses a moment during a quiet overnight period and starts a timer. The analyst then records how much time passes before the next security alert is generated. For example, if monitoring begins at 1:10 a.m. and the next alert is generated at 1:17 a.m., the recorded time is 7 minutes.

Question.

C4.1 Identify the random variable and state the possible values it can take.

C4.1 - Identify the random variable and its possible values

Let

$$T = \text{the time from a randomly selected observation time until the next security alert is generated}$$

The random variable T measures elapsed time, so it cannot take negative values. Therefore,

$$T \geq 0 \ .$$

1.4 Battery life of mobile devices

A mobile phone company in Bogotá tests the batteries used in some of its devices. As part of quality control, the company monitors how long each battery continues to operate before it fails.

An analyst selects a battery at random and records the total operating time from the beginning of the test until the battery stops working. For example, one battery might operate for 18.5 hours before failing, while another might operate for 23.2 hours.

Question.

C4.1 Identify the random variable and state the possible values it can take.

C4.1 - Identify the random variable and its possible values

Let

T = the operating time of a randomly selected battery until it f

The random variable T measures elapsed operating time, so it cannot take negative values. Therefore,

$$T \geq 0 \, .$$

2 Finding Probabilities from an Exponential Distribution

Problems:

2.1. Customer-support calls

2.2. Food-delivery requests

2.3. Bank security alerts

2.4. Battery failures

2.5. Ride-hailing requests

2.6. Food-delivery preparation times

2.1 Customer-support calls

A mobile phone company in Bogotá receives technical-support calls throughout the afternoon. The time, in minutes, from a randomly selected observation time until the next support call is modeled by an exponential distribution with rate 2 calls per minute.

Question.

C4 Find the probability that the next support call takes more than 5 minutes to arrive.

Define the random variable

$T =$ waiting time until the next support call, in minutes

C4.3 - Find a probability

The company wants to find the probability that the next support call takes more than 5 minutes to arrive, i.e.

$$P(T > 5)$$

General	$P(T > t) = e^{-\lambda t}$
Context	$P(T > 5), \quad \lambda = 2$

Therefore,

$$P(T > 5) = e^{-(2)(5)} = e^{-10}$$

$$P(T > 5) \approx 0.0000454$$

2.2 Food-delivery requests

A food-delivery platform in Bogotá receives delivery requests throughout the evening. The time, in minutes, from a randomly selected observation time until the next delivery request is modeled by an exponential distribution with rate 0.15 requests per minute.

Question.

C4 Find the probability that the next delivery request takes less than 10 minutes to arrive.

Define the random variable

$T =$ waiting time until the next delivery request, in minutes

C4.3 - Find a probability

The platform wants to find the probability that the next delivery request takes less than 10 minutes to arrive, i.e.

$$P(T < 10)$$

General	$P(T < t) = 1 - e^{-\lambda t}$
Context	$P(T < 10), \quad \lambda = 0.15$

Therefore,

$$P(T < 10) = 1 - e^{-(0.15)(10)} = 1 - e^{-1.5}$$

$$P(T < 10) \approx 0.7769$$

2.3 Bank security alerts

A Colombian digital bank continuously monitors transactions for unusual activity. The time, in minutes, from a randomly selected observation time until the next security alert is modeled by an exponential distribution with an average of 1 alert every 10 minutes.

Question.

C4 Find the probability that the next security alert takes more than 8 minutes to be generated.

Define the random variable

$T =$ waiting time until the next security alert, in minutes

C4.2 - Find the rate parameter

The average rate is 1 alert every 10 minutes, so

$$\lambda = \frac{1}{10} = 0.10 \text{ alerts per minute}$$

C4.3 - Find a probability

The bank wants to find the probability that the next security alert takes more than 8 minutes to be generated, i.e.

$$P(T > 8)$$

General	$P(T > t) = e^{-\lambda t}$
Context	$P(T > 8), \quad \lambda = 0.10$

Therefore,

$$P(T > 8) = e^{-(0.10)(8)} = e^{-0.8}$$

$$P(T > 8) \approx 0.4493$$

2.4 Battery failures

A mobile phone company tests the batteries used in its devices. The operating time, in hours, of a randomly selected battery before it fails is modeled by an exponential distribution with an average of 1 failure every 25 hours.

Question.

C4 Find the probability that a randomly selected battery fails within its first 15 hours of operation.

Define the random variable

$T =$ operating time of a randomly selected battery, in hours

C4.2 - Find the rate parameter

The average rate is 1 failure every 25 hours, so

$$\lambda = \frac{1}{25} = 0.04 \text{ failures per hour}$$

C4.3 - Find a probability

The company wants to find the probability that a randomly selected battery fails within its first 15 hours of operation, i.e.

$$P(T < 15)$$

General	$P(T < t) = 1 - e^{-\lambda t}$
Context	$P(T < 15), \quad \lambda = 0.04$

Therefore,

$$P(T < 15) = 1 - e^{-(0.04)(15)} = 1 - e^{-0.6}$$

$$P(T < 15) \approx 0.4512$$

2.5 Ride-hailing requests

A ride-hailing platform in Bogotá receives ride requests throughout the day. The time, in minutes, from a randomly selected observation time until the next ride request is modeled by an exponential distribution with rate 0.20 requests per minute.

Question.

C4 Find the probability that the next ride request arrives between 3 and 8 minutes after the observation time.

Define the random variable

$T =$ waiting time until the next ride request, in minutes

C4.3 - Find a probability

The platform wants to find the probability that the next ride request arrives between 3 and 8 minutes after the observation time, i.e.

$$P(3 < T < 8)$$

General	$P(a < T < b) = e^{-\lambda a} - e^{-\lambda b}$
Context	$P(3 < T < 8), \quad \lambda = 0.20$

Therefore,

$$P(3 < T < 8) = \left(1 - e^{-(0.20)(8)}\right) - \left(1 - e^{-(0.20)(3)}\right)$$

$$= e^{-0.6} - e^{-1.6}$$

$$P(3 < T < 8) \approx 0.3672$$

2.6 Food-delivery preparation times

A food-delivery restaurant in Bogotá prepares orders throughout the evening. The preparation time, in minutes, for a randomly selected order is modeled by an exponential distribution with an average of 1 order every 12 minutes.

Question.

C4 Find the probability that a randomly selected order takes between 5 and 15 minutes to prepare.

Define the random variable

$T =$ preparation time of a randomly selected order, in minutes

C4.2 - Find the rate parameter

The average rate is 1 order every 12 minutes, so

$$\lambda = \frac{1}{12} \approx 0.0833 \text{ orders per minute}$$

C4.3 - Find a probability

The restaurant wants to find the probability that an order takes between 5 and 15 minutes to prepare, i.e.

$$P(5 < T < 15)$$

General	$P(a < T < b) = e^{-\lambda a} - e^{-\lambda b}$
Context	$P(5 < T < 15), \quad \lambda = \frac{1}{12}$

Therefore,

$$P(5 < T < 15) = \left(1 - e^{-(1/12)(15)}\right) - \left(1 - e^{-(1/12)(5)}\right)$$

$$= e^{-5/12} - e^{-15/12}$$

$$P(5 < T < 15) \approx 0.3413$$

3 Finding the Rate of an Exponential Distribution

Problems:

3.1. Customer-support calls

3.2. Food-delivery requests

3.3. Bank security alerts

3.4. Battery failures

3.5. Ride-hailing requests

3.6. Food-delivery requests (interval)

3.1 Customer-support calls

A mobile phone company in Bogotá receives technical-support calls throughout the afternoon. The time T, in minutes, between a randomly selected observation time and the next support call is modeled by an exponential distribution.

Based on previous observations, the company estimates that there is a 90% probability that the next support call arrives within 10 minutes.

Question.

C4.2 Find the rate parameter λ of the exponential distribution.

C4.2 - Find the rate parameter

General	$P(T \leq t) = 1 - e^{-\lambda t}$
Context	$P(T \leq 10) = 0.90$

Therefore,

$$0.90 = 1 - e^{-10\lambda} \quad \longrightarrow \quad e^{-10\lambda} = 0.10$$

$$\longrightarrow \quad \ln(e^{-10\lambda}) = \ln(0.10)$$

$$\longrightarrow \quad -10\lambda = \ln(0.10) \quad \longrightarrow \quad \lambda = -\frac{\ln(0.10)}{10}$$

$\lambda \approx 0.2303$ calls/minute

3.2 Food-delivery requests

A food-delivery platform in Bogotá receives delivery requests throughout the evening. The time T, in minutes, from a randomly selected observation time until the next delivery request is modeled by an exponential distribution.

Based on previous observations, the platform estimates that there is a 75% probability that the next delivery request arrives within 8 minutes.

Question.

C4.2 Find the rate parameter λ of the exponential distribution.

C4.2 - Find the rate parameter

General	$P(T \leq t) = 1 - e^{-\lambda t}$
Context	$P(T \leq 8) = 0.75$

Therefore,

$$0.75 = 1 - e^{-8\lambda} \quad \longrightarrow \quad e^{-8\lambda} = 0.25$$

$$\longrightarrow \quad \ln(e^{-8\lambda}) = \ln(0.25)$$

$$\longrightarrow \quad -8\lambda = \ln(0.25) \quad \longrightarrow \quad \lambda = -\frac{\ln(0.25)}{8}$$

$\lambda \approx 0.1733$ requests/minute

3.3 Bank security alerts

A Colombian digital bank continuously monitors transactions for unusual activity. The time T, in minutes, from a randomly selected observation time until the next security alert is modeled by an exponential distribution.

During a quiet overnight period, the bank estimates that there is a 20% probability that the next security alert takes more than 12 minutes to be generated.

Question.

C4.2 Find the rate parameter λ of the exponential distribution.

C4.2 - Find the rate parameter

General	$P(T \geq t) = e^{-\lambda t}$
Context	$P(T \geq 12) = 0.20$

Therefore,

$$0.20 = e^{-12\lambda}$$

$$\longrightarrow \quad \ln(e^{-12\lambda}) = \ln(0.20)$$

$$\longrightarrow \quad -12\lambda = \ln(0.20) \quad \longrightarrow \quad \lambda = -\frac{\ln(0.20)}{12}$$

$$\lambda \approx 0.1341 \text{ alerts/minute}$$

3.4 Battery failures

A company tests rechargeable batteries used in mobile devices. The operating time T, in hours, until a randomly selected battery fails is modeled by an exponential distribution.

Based on previous tests, the company estimates that there is a 30% probability that a battery operates for at least 15 hours before failing.

Question.

C4.2 Find the rate parameter λ of the exponential distribution.

C4.2 - Find the rate parameter

General	$P(T \geq t) = e^{-\lambda t}$
Context	$P(T \geq 15) = 0.30$

Therefore,

$$0.30 = e^{-15\lambda}$$

$$\longrightarrow \quad \ln(e^{-15\lambda}) = \ln(0.30)$$

$$\longrightarrow \quad -15\lambda = \ln(0.30) \quad \longrightarrow \quad \lambda = -\frac{\ln(0.30)}{15}$$

$$\lambda \approx 0.0803 \text{ failures/hour}$$

3.5 Ride-hailing requests

A ride-hailing platform in Bogotá records the time T, in minutes, from a randomly selected observation time until the next ride request. The waiting time is modeled by an exponential distribution.

Based on previous observations, the platform estimates that there is a 40% probability that the next ride request arrives between 3 and 8 minutes after the observation time.

Question.

C4.2 Find the rate parameter λ of the exponential distribution.

C4.2 - Find the rate parameter

General	$P(a < T < b) = e^{-\lambda a} - e^{-\lambda b}$
Context	$P(3 < T < 8) = 0.40$

Therefore,

$$0.40 = e^{-3\lambda} - e^{-8\lambda}$$

Since λ appears in two exponential terms, the equation must be solved numerically.

$$\lambda \approx 0.1006 \text{ requests/minute}$$

3.6 Food-delivery requests

A food-delivery platform in Bogotá receives requests throughout the evening. The time T, in minutes, from a randomly selected observation time until the next delivery request is modeled by an exponential distribution.

Based on previous observations, the platform estimates that there is a 30% probability that the next delivery request arrives between 5 and 15 minutes after the observation time.

Question.

C4.2 Find the rate parameter λ of the exponential distribution.

C4.2 - Find the rate parameter

General	$P(a < T < b) = e^{-\lambda a} - e^{-\lambda b}$
Context	$P(5 < T < 15) = 0.30$

Therefore,

$$0.30 = e^{-5\lambda} - e^{-15\lambda}$$

Since λ appears in two exponential terms, the equation must be solved numerically.

$$\lambda \approx 0.0696 \text{ requests/minute}$$